

TABLES

CYBER SECURITY ISSUES IN INDIA” -- INDIA’S APPROACH TO CYBER SECURITY AND NATIONAL CYBER SECURITY

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Table I: MAJOR CYBERSECURITY CHALLENGES AND THEIR IMPACT

S. No.	Challenge	% Affected / Share	Annual Growth (%)	Severity Score (1–5)
1	Lack of Awareness	65%	10%	4
2	Skilled Workforce Gap	35%	20%	4
3	Rising Cyber Attacks	100% (1.3M cases)	18%	5
4	Low Conviction Rate	28%	5%	4
5	Digital Expansion Risk	80% (800M users)	9%	5
6	Data Breach Incidents	20%	15%	4

7	SME Security Gap	55%	6%	3
8	Mobile-Based Attacks	72%	11%	4
9	Insider Threats	22%	4%	4
10	Phishing Attacks	38%	15%	5

TABLE II YEAR-WISE CYBER SECURITY INCIDENTS IN INDIA (2018–2025)
— SOURCE: CERT-IN / LOK SABHA / PIB

Year	Cyber Incidents Reported	YoY Growth (%)	Source
2018	2,08,456	—	CERT-In
2019	3,94,499	+89.2%	CERT-In
2020	11,58,208	+193.6%	CERT-In
2021	14,02,809	+21.1%	CERT-In
2022	13,91,457	-0.8%	CERT-In
2023	15,92,917	+14.5%	MeitY/Lok Sabha
2024*	22,68,000*	+42.4%*	MHA/PIB
2025†	29,44,000	+29.8%	CERT-In/PIB Jan 2026

**TABLE III DISTRIBUTION OF CYBER ATTACK TYPES IN INDIA (2023) —
SOURCE: CERT-IN / I4C / DSCI**

Attack Type	Share of Incidents (%)	Primary Target	Severity
Financial Fraud & Phishing	60–65%	Individuals / Banks	Very High
Malware & Ransomware	12–15%	Enterprises / SMEs	High
Unauthorized Network Scanning	8–10%	Govt. / Critical Infra.	High
DDoS Attacks	5–7%	Web Services	Medium
Data Breaches	5–6%	Healthcare / Finance	Very High
Social Engineering	4–5%	Users / Employees	High
Cyber Espionage / APT	2–3%	Government / Defence	Critical

**TABLE IV CONSOLIDATED QUANTITATIVE FINDINGS — INDIA'S CYBER SECURITY
LANDSCAPE (2023–2025)**

Parameter / Finding	Metric / Value	Annual Growth Rate	Source
Cyber incidents reported (2023)	15,92,917	+14.5%	CERT-In [3]
Cyber incidents reported (2024)*	22,68,000*	+42.4%	MHA [4]
Cyber fraud financial losses (2024)	₹22,845 Cr.	+206%	I4C / PIB [4]
Cyber incidents reported (2025)†	29,44,000	+29.8%	CERT-In/PIB [21]
Cyber fraud financial losses (2025)†	₹22,495 Cr.	–1.5%	MHA/PIB [21]

Parameter / Finding	Metric / Value	Annual Growth Rate	Source
Avg. data breach cost — India	\$2.18 million	+28% (3 yrs)	IBM 2024 [7]
Internet users in India	100.29 Cr (1,002.9M)	+8% p.a.	TRAI 2025 [8]
Mobile internet share	~70–75%	—	GSMA 2025 [9]
Human element in breaches (global)	68%	—	Verizon [2]
Users lacking cyber hygiene	~65%	—	Survey / [2]
Phishing-attributed incidents	35–40%	—	Verizon [2]
Cybersecurity workforce shortage	~800,000 professionals	—	WEF 2024 [10]
SMEs lacking cyber infrastructure	~50–60%	—	FICCI [11]
MFA / Backup effectiveness	>90%	—	NIST [12]
Antivirus / Firewall effectiveness	80–85%	—	NIST [12]
Advanced security adoption rate	~25–30%	—	Cisco [13]
Orgs. at progressive+ readiness	41%	—	Cisco [13]

TABLE V EFFECTIVENESS VS. ADOPTION RATE OF CYBER SECURITY TECHNIQUES IN INDIA

Security Technique	Effectiveness (%)	Adoption Rate — India (%)	Implementation Gap
Multi-Factor Authentication (MFA)	>92%	~28%	Very High
Data Backup & Recovery	>90%	~35%	High
Encryption (End-to-End)	~88%	~40%	High
IDS / IPS Systems	~85%	~25%	Very High
Firewalls	~83%	~65%	Moderate
Antivirus / Anti-malware	~80%	~70%	Low
User Awareness Training	~75%	~20%	Very High
Regular Software Patching	~85%	~45%	High